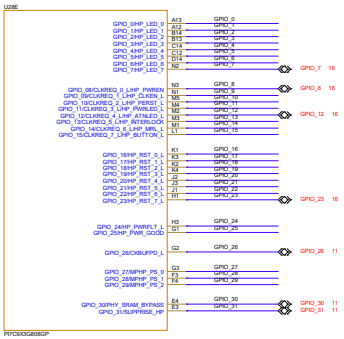
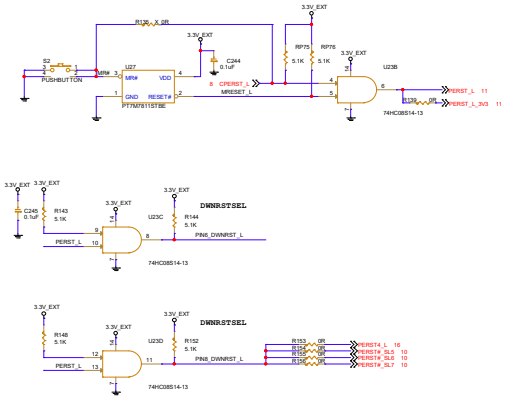
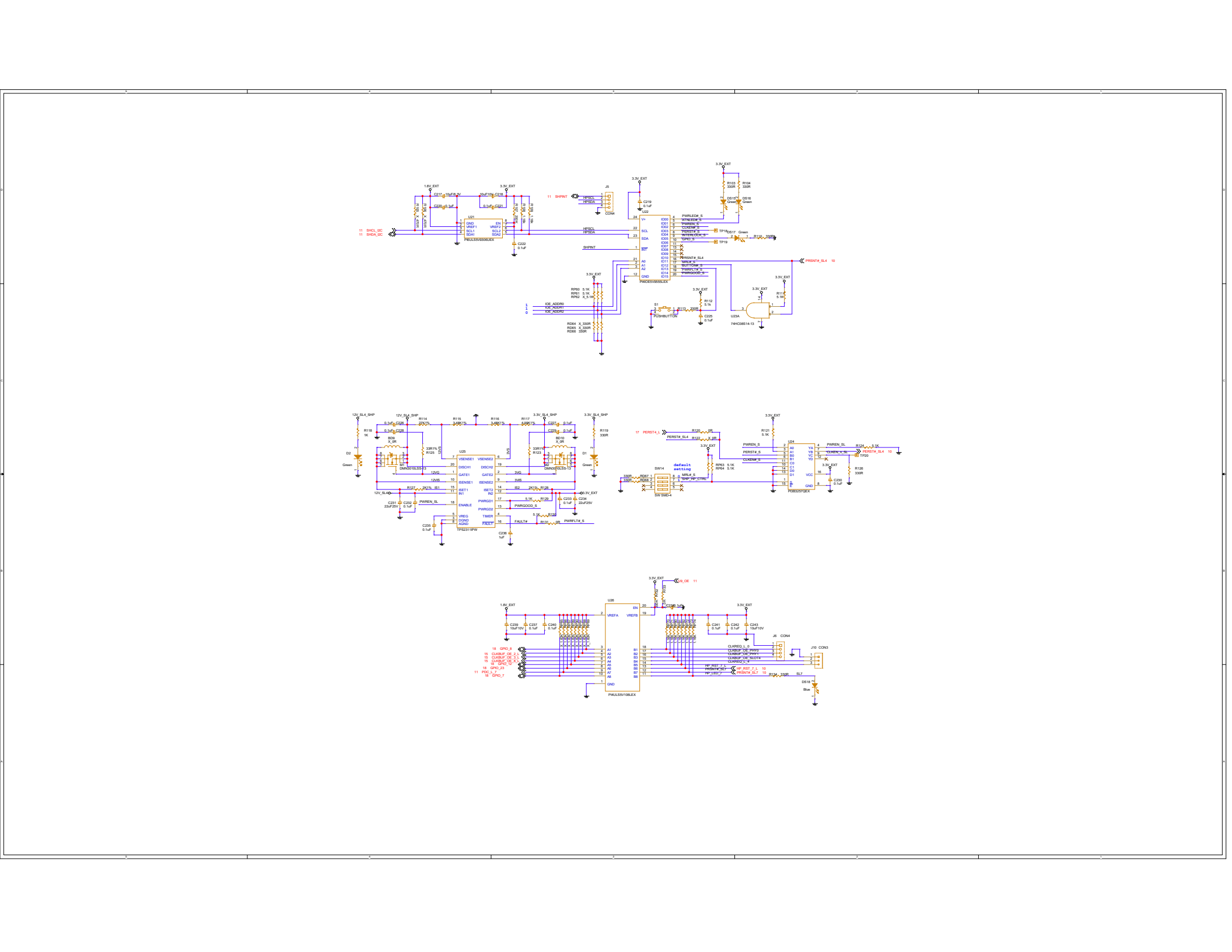
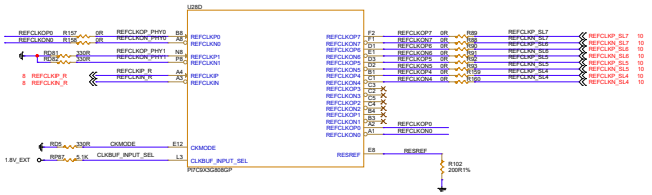
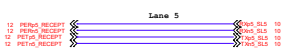
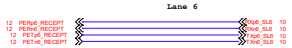

U29E
GPIO_0/HP_LED_0
GPIO_1/HP_LED_1
GPIO_2/HP_LED_2
GPIO_3/HP_LED_3
GPIO_4/HP_LED_4
GPIO_5/HP_LED_5
GPIO_6/HP_LED_6
GPIO_7/HP_LED_7
GPIO_08/CLKREQ_0_L/HP_PWREN
GPIO_09/CLKREQ_1_L/HP_CLKEN_L
GPIO_10/CLKREQ_2_L/HP_PERST_L
GPIO_11/CLKREQ_3_L/HP_PWRLED_L
GPIO_12/CLKREQ_4_L/HP_ATNLED_L
GPIO_13/CLKREQ_5_L/HP_INTERLOCK
GPIO_14/CLKREQ_6_L/HP_MRL_L
GPIO_15/CLKREQ_7_L/HP_BUTTON_L
GPIO_16/HP_RST_0_L
GPIO_17/HP_RST_1_L
GPIO_18/HP_RST_2_L
GPIO_19/HP_RST_3_L
GPIO_20/HP_RST_4_L
GPIO_21/HP_RST_5_L
GPIO_22/HP_RST_6_L
GPIO_23/HP_RST_7_L
GPIO_24/HP_PWRFLT_L
GPIO_25/HP_PWR_GOOD
GPIO_26/CK8LIFPD_L
GPIO_27/MPHP_PS_0
GPIO_28/MPHP_PS_1
GPIO_29/MPHP_PS_2
GPIO_30/PHY_SRAM_BYPASS
GPIO_31/SUPPRSE_HP
A13
A12
B14
B13
C14
C13
D12
N2
N3
N1
M5
M4
M2
M3
M1
L1
K1
K3
K2
K4
J2
J3
J1
H1
H3
G1
G2
G3
F3
F4
E4
E3
GPIO_0
GPIO_1
GPIO_2
GPIO_3
GPIO_4
GPIO_5
GPIO_6
GPIO_7
GPIO_8
GPIO_9
GPIO_10
GPIO_11
GPIO_12
GPIO_13
GPIO_14
GPIO_15
GPIO_16
GPIO_17
GPIO_18
GPIO_19
GPIO_20
GPIO_21
GPIO_22
GPIO_23
GPIO_24
GPIO_25
GPIO_26
GPIO_27
GPIO_28
GPIO_29
GPIO_30
GPIO_31
GPIO_7 16
GPIO_8 16
GPIO_12 16
GPIO_23 16
GPIO_26 11
GPIO_30 11
GPIO_31 11
P17C30X3G803BGP


3.3V_EXT
R136 X_0R
S2
PUSHBUTTON
MR#
U27
MR#
VDD
GND
RESET#
PT7M7811STBE
C244
0.1uF
8 CPERST_L
MRESET_L
RP75
RP76
5.1K
5.1K
U23B
74HC08S14-13
R139 0R
PERST_L 11
PERST_L_3V3 11
C245
0.1uF
R143
5.1K
U23C
R144
5.1K
DWNRSTSEL
PERST_L
PIN6_DWNRST_L
R148
5.1K
U23D
R152
5.1K
R153 0R
R154 0R
R155 0R
R156 0R
PERST4_L 16
PERST4_SL5 10
PERST4_SL6 10
PERST4_SL7 10

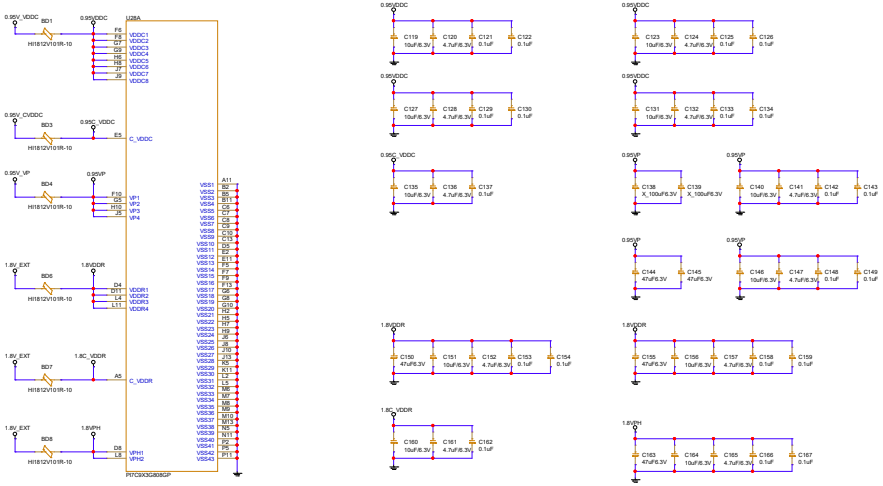


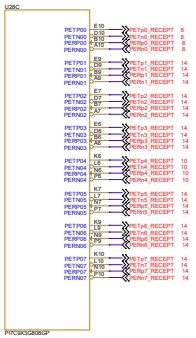

U28D
REFCLKOP0
REFCLKON0
R157
0R
R158
0R
REFCLKOP_PHY0
REFCLKON_PHY0
REFCLKOP_PHY1
REFCLKON_PHY1
RD41
RD42
330R
330R
REFCLKP_R
REFCLKN_R
8 REFCLKP_R
9 REFCLKN_R
B8
A8
B9
A9
A4
A3
REFCLKP0
REFCLKN0
REFCLKP1
REFCLKN1
REFCLKP
REFCLKN
RD5
33ΩR
CKMODE
E12
CKMODE
RP87
5.1K
CLKBUF_INPUT_SEL
L3
CLKBUF_INPUT_SEL
1.8V_EXT
PITCXC3G30BCP
REFCLKOP7
REFCLKON7
REFCLKOP6
REFCLKON6
REFCLKOP5
REFCLKON5
REFCLKOP4
REFCLKON4
REFCLKOP3
REFCLKON3
REFCLKOP2
REFCLKON2
REFCLKOP1
REFCLKON1
REFCLKOP0
REFCLKON0
F2
F1
D1
E1
D3
D2
B1
C1
C2
C3
B2
B3
A2
A1
REFCLKOP0
REFCLKGND
E8
RESREF
RESREF
R102
200R1%
F2
F1
D1
E1
D3
D2
B1
C1
REFCLKOP7
REFCLKON7
REFCLKOP6
REFCLKON6
REFCLKOP5
REFCLKON5
REFCLKOP4
REFCLKON4
REFCLKGND
R99
R98
R100
R101
R102
R103
R104
R105
0R
0R
0R
0R
0R
0R
0R
0R
REFCLKP_SL7
REFCLKN_SL7
REFCLKP_SL6
REFCLKN_SL6
REFCLKP_SL5
REFCLKN_SL5
REFCLKP_SL4
REFCLKN_SL4
REFCLKP_SL7
REFCLKN_SL7
REFCLKP_SL6
REFCLKN_SL6
REFCLKP_SL5
REFCLKN_SL5
REFCLKP_SL4
REFCLKN_SL4
10
10
10
10
10
10
10
10

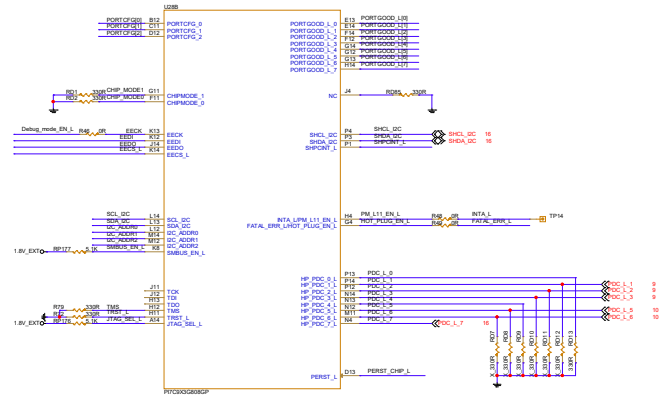

Lane 5
12 PERp5_RECEPT
12 PERn5_RECEPT
12 PETp5_RECEPT
12 PETn5_RECEPT
TXp5_SL5 10
TXn5_SL5 10
TXp5_SL5 10
TXn5_SL5 10

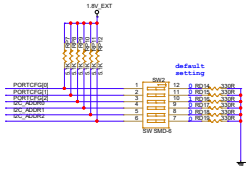

Lane 6
12 PERp6_RECEPT
12 PERn6_RECEPT
12 PETp6_RECEPT
12 PETn6_RECEPT
TXp6_SL6 10
TXn6_SL6 10
TXp6_SL6 10
TXn6_SL6 10


Lane 7
12 PETp7_RECEPT
12 PETn7_RECEPT
12 PERp7_RECEPT
12 PERn7_RECEPT
TXp7_SL7 10
TXn7_SL7 10
RXp7_SL7 10
RXn7_SL7 10

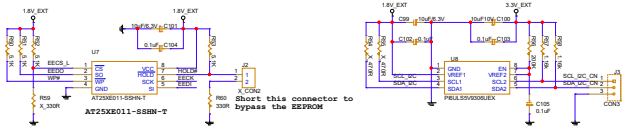

0.95V_VDDC
BD1
0.95VDDC
H1812V101R-10
F6
F8
G7
G9
H6
H8
J7
J9
VDDC1
VDDC2
VDDC3
VDDC4
VDDC5
VDDC6
VDDC7
VDDC8
0.95V_CVDDC
BD3
0.95C_VDDC
E5
C_VDDC
H1812V101R-10
0.95V_VP
BD4
0.95VP
F10
G5
H10
J5
VP1
VP2
VP3
VP4
H1812V101R-10
1.8V_EXT
BD6
1.8VDDR
D4
D11
L4
L11
VDDR1
VDDR2
VDDR3
VDDR4
H1812V101R-10
1.8V_EXT
BD7
1.8C_VDDR
A5
C_VDDR
H1812V101R-10
1.8V_EXT
BD8
1.8VPH
D8
L8
VPH1
VPH2
H1812V101R-10
U28A
VSS1
VSS2
VSS3
VSS4
VSS56
VSS57
VSS58
VSS59
VSS10
VSS11
VSS12
VSS13
VSS14
VSS15
VSS16
VSS17
VSS18
VSS19
VSS20
VSS21
VSS22
VSS23
VSS24
VSS25
VSS26
VSS27
VSS28
VSS29
VSS30
VSS31
VSS32
VSS33
VSS34
VSS35
VSS36
VSS37
VSS38
VSS39
VSS40
VSS41
VSS42
A11
B2
B6
B11
C5
C7
C8
C9
C10
C13
D5
E2
E11
F1
F2
F4
F13
G8
G9
G10
H2
H5
H7
H9
J6
J8
J10
J13
K5
K11
L1
L5
M6
M7
M8
M9
M10
M13
N5
N11
P2
P5
P11
PI7C9X3G808GP
0.95VDDC
C119
10uF/6.3V
C120
4.7uF/6.3V
C121
0.1uF
C122
0.1uF
0.95VDDC
C123
10uF/6.3V
C124
4.7uF/6.3V
C125
0.1uF
C126
0.1uF
0.95VDDC
C127
10uF/6.3V
C128
4.7uF/6.3V
C129
0.1uF
C130
0.1uF
0.95VDDC
C131
10uF/6.3V
C132
4.7uF/6.3V
C133
0.1uF
C134
0.1uF
0.95C_VDDC
C135
10uF/6.3V
C136
4.7uF/6.3V
C137
0.1uF
0.95VP
C138
X_100uF/6.3V
C139
X_100uF/6.3V
0.95VP
C140
10uF/6.3V
C141
4.7uF/6.3V
C142
0.1uF
C143
0.1uF
0.95VP
C144
47uF/6.3V
C145
47uF/6.3V
0.95VP
C146
10uF/6.3V
C147
4.7uF/6.3V
C148
0.1uF
C149
0.1uF
1.8VDDR
C150
47uF/6.3V
C151
10uF/6.3V
C152
4.7uF/6.3V
C153
0.1uF
C154
0.1uF
1.8VDDR
C155
47uF/6.3V
C156
10uF/6.3V
C157
4.7uF/6.3V
C158
0.1uF
C159
0.1uF
1.8C_VDDR
C160
10uF/6.3V
C161
4.7uF/6.3V
C162
0.1uF
1.8VPH
C163
47uF/6.3V
C164
10uF/6.3V
C165
4.7uF/6.3V
C166
0.1uF
C167
0.1uF

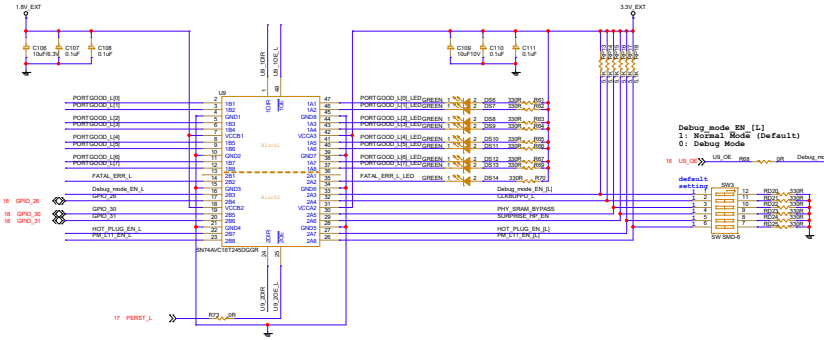

U28C
PETP00
PETN00
PERP00
PERN00
E10
D10
B10
A10
PETP01
PETN01
PERP01
PERN01
E9
D9
B9
A9
PETP02
PETN02
PERP02
PERN02
E7
D7
B7
A7
PETP03
PETN03
PERP03
PERN03
E6
D6
B6
A6
PETP04
PETN04
PERP04
PERN04
K8
L8
N8
P8
PETP05
PETN05
PERP05
PERN05
K7
L7
N7
P7
PETP06
PETN06
PERP06
PERN06
K9
L9
N9
P9
PETP07
PETN07
PERP07
PERN07
K10
L10
N10
P10
PETp0_RECEPT 8
PETn0_RECEPT 8
PERp0_RECEPT 8
PERn0_RECEPT 8
PETp1_RECEPT 14
PETn1_RECEPT 14
PERp1_RECEPT 14
PERn1_RECEPT 14
PETp2_RECEPT 14
PETn2_RECEPT 14
PERp2_RECEPT 14
PERn2_RECEPT 14
PETp3_RECEPT 14
PETn3_RECEPT 14
PERp3_RECEPT 14
PERn3_RECEPT 14
PETp4_RECEPT 10
PETn4_RECEPT 10
PERp4_RECEPT 10
PERn4_RECEPT 10
PETp5_RECEPT 14
PETn5_RECEPT 14
PERp5_RECEPT 14
PERn5_RECEPT 14
PETp6_RECEPT 14
PETn6_RECEPT 14
PERp6_RECEPT 14
PERn6_RECEPT 14
PETp7_RECEPT 14
PETn7_RECEPT 14
PERp7_RECEPT 14
PERn7_RECEPT 14
PITC9X3G80BGP

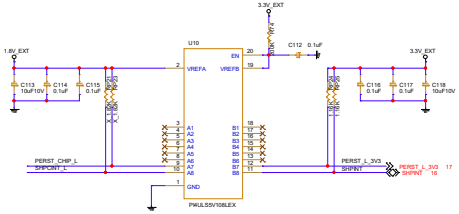


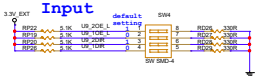


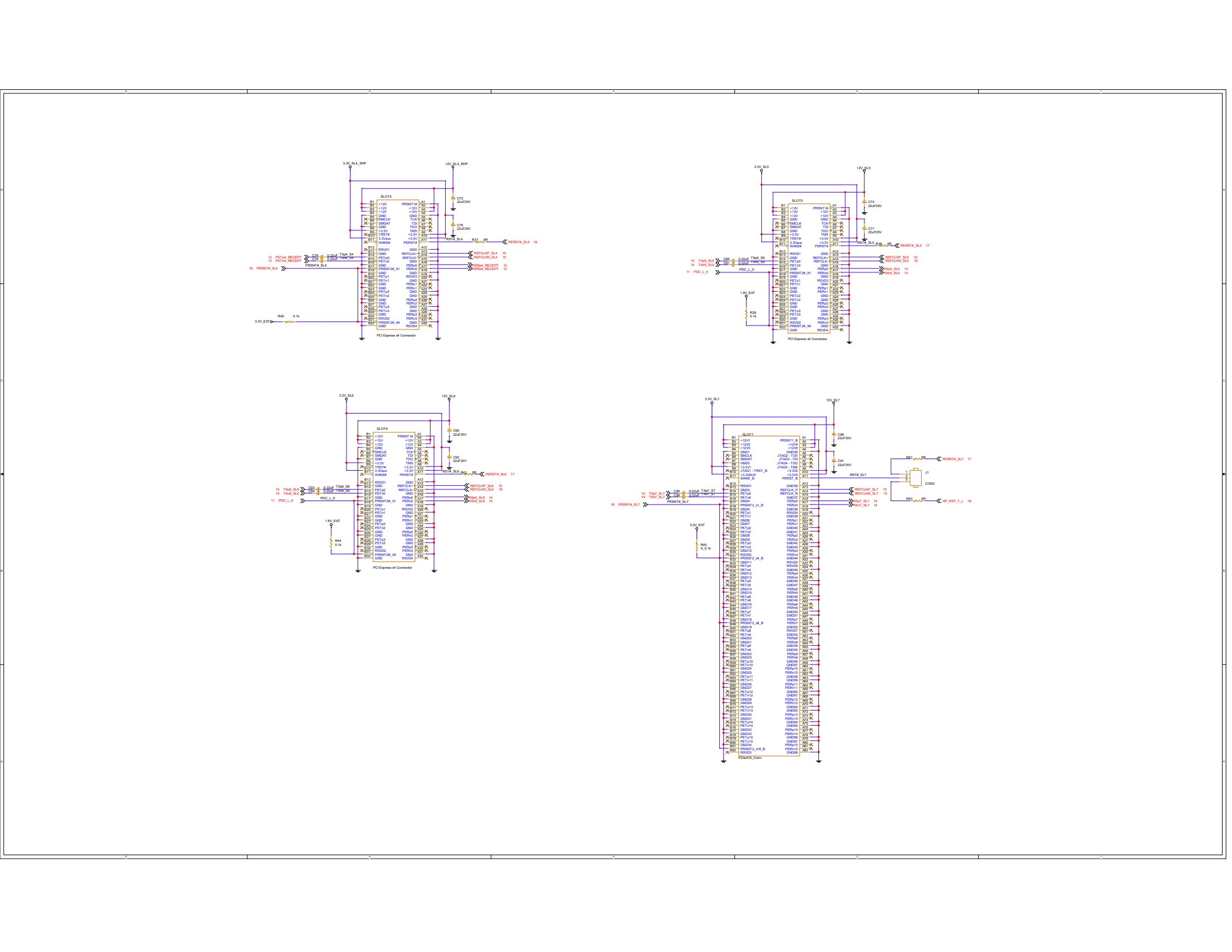
PORTCFG[2]	PORTCFG[1]	PORTCFG[0]	Functional Mode
0	0	0	Reserved
0	0	1	2Port - 8Lane Configuration
0	1	0	3Port - 8Lane Configuration
0	1	1	4Port - 8Lane Configuration
1	0	0	5Port - 8Lane Configuration
1	0	1	8Port - 8Lane Configuration
1	1	0	Reserved
1	1	1	Reserved

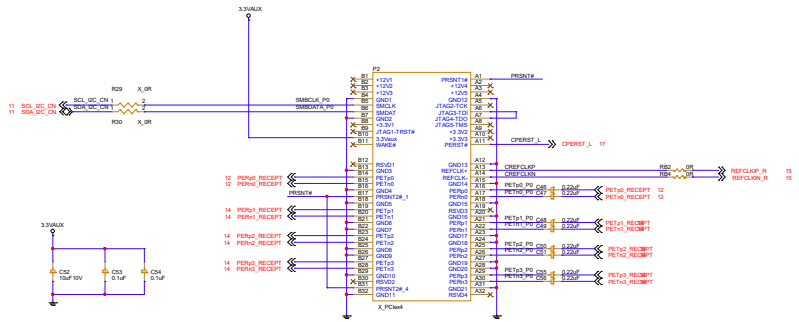


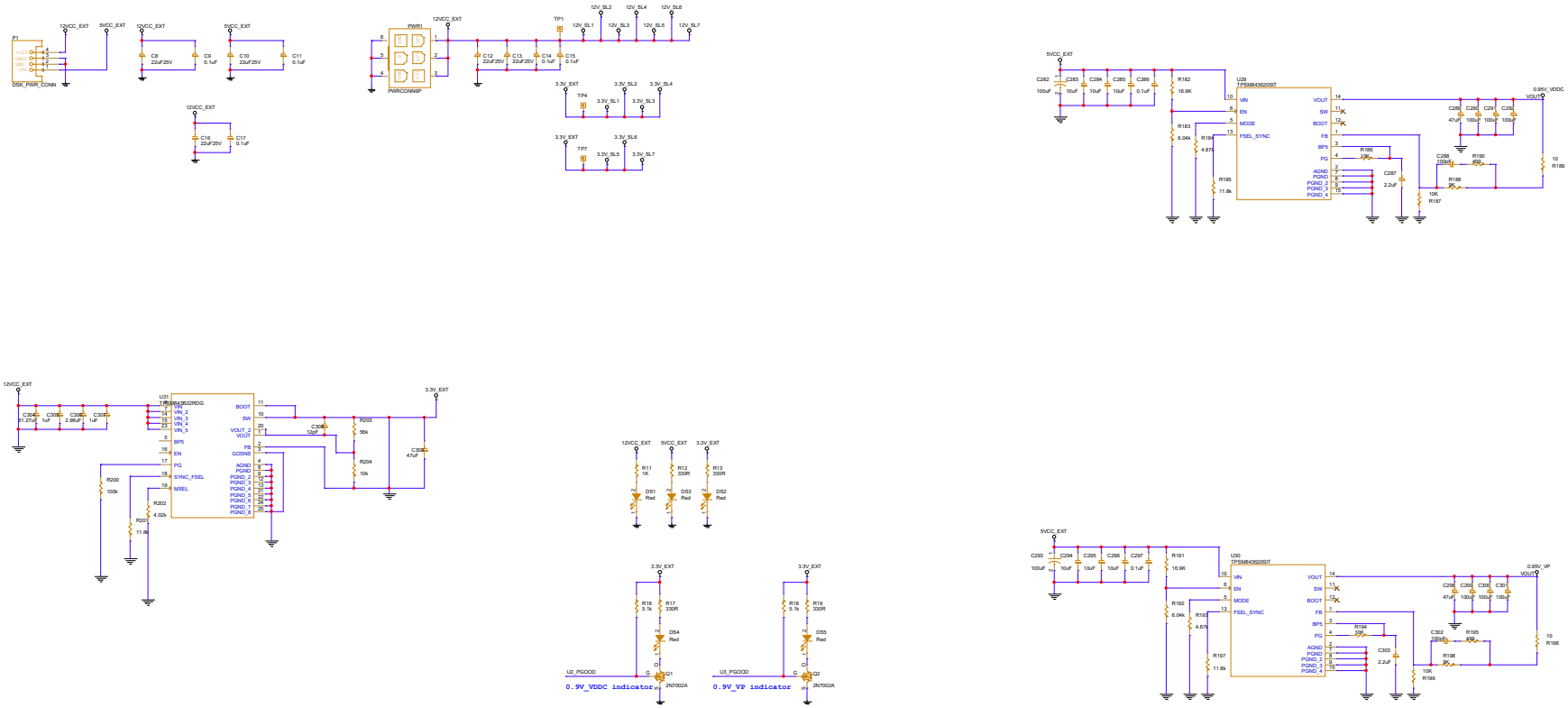


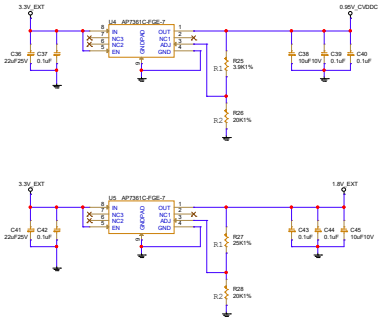





3.3V_SL4_SHP
12V_SL4_SHP
SLOT4
C75 22uF/25V
C76 22uF/25V
PRSNT#_SL4
RST#_SL4
R37 0R
PERST#_SL4 16
PCI Express x4 Connector
C78 0.22uF TXp4_S4
C79 0.22uF TXn4_S4
12 PEDp4_RECEPIT
12 PEDn4_RECEPIT
16 PRSNT#_SL4
R40 5.1k
3.3V_EXT
REFCLKP_SL4 15
REFCLKN_SL4 15
PERp4_RECEPIT 12
PERn4_RECEPIT 12
3.3V_SL5
12V_SL5
SLOT5
C74 22uF/25V
C77 22uF/25V
RST#_SL5
R38 0R
PERST#_SL5 17
C80 0.22uF TXp5_S5
C81 0.22uF TXn5_S5
14 TXp5_SL5
14 TXn5_SL5
11 PDC_L_5
1.8V_EXT
R39 5.1k
REFCLKP_SL5 15
REFCLKN_SL5 15
PERp5_SL5 14
PERn5_SL5 14
3.3V_SL6
12V_SL6
SLOT6
C90 22uF/25V
C92 22uF/25V
RST#_SL6
R43 0R
PERST#_SL6 17
C93 0.22uF TXp6_S6
C94 0.22uF TXn6_S6
14 TXp6_SL6
14 TXn6_SL6
11 PDC_L_6
1.8V_EXT
R44 5.1k
REFCLKP_SL6 15
REFCLKN_SL6 15
PERp6_SL6 14
PERn6_SL6 14
3.3V_SL7
12V_SL7
SLOT7
C89 22uF/25V
C91 22uF/25V
RST#_SL7
R41 0R
PERST#_SL7 17
J1
CON3
R43 0R
HP_RST_7_L 16
C95 0.22uF TXp7_S7
C96 0.22uF TXn7_S7
14 TXp7_SL7
14 TXn7_SL7
PRSNT#_SL7
16 PRSNT#_SL7
3.3V_EXT
R45 X_5.1k
REFCLKP_SL7 15
REFCLKN_SL7 15
PERp7_SL7 14
PERn7_SL7 14
PCIeX16_Conn


3.3VAUX
P2
R29 X_0R
R30 X_0R
SCL_I2C_CN
SDA_I2C_CN
SMBCLK_P0
SMBDATA_P0
PRSNT#
CPERST_L
CREFCLKP
CREFCLKN
REFCLKP_R
REFCLKN_R
PETp0_P0
PETn0_P0
PETp1_P0
PETn1_P0
PETp2_P0
PETn2_P0
PETp3_P0
PETn3_P0
0.22uF
PERp0_RECEPT
PERn0_RECEPT
PERp1_RECEPT
PERn1_RECEPT
PERp2_RECEPT
PERn2_RECEPT
PERp3_RECEPT
PERn3_RECEPT
PETp0_RECEPT
PETn0_RECEPT
PETp1_RECEPT
PETn1_RECEPT
PETp2_RECEPT
PETn2_RECEPT
PETp3_RECEPT
PETn3_RECEPT
3.3VAUX
C52 10uF 10V
C53 0.1uF
C54 0.1uF
X_PCIex4


P1
DSK_PWR_CONN
12VCC_EXT
5VCC_EXT
C8 22uF/25V
C9 0.1uF
C10 22uF/25V
C11 0.1uF
C16 22uF/25V
C17 0.1uF
PWR1
PWRCONN6P
C12 22uF/25V
C13 22uF/25V
C14 0.1uF
C15 0.1uF
TP1
12V_SL1
12V_SL2
12V_SL3
12V_SL4
12V_SL5
12V_SL6
12V_SL7
3.3V_EXT
TP4
3.3V_SL1
3.3V_SL2
3.3V_SL3
3.3V_SL4
TP7
3.3V_SL5
3.3V_SL6
3.3V_SL7
U29
TPSM843620SIT
VIN
EN
MODE
FSEL_SYNC
VOUT
SW
BOOT
FB
BPS
PG
AGND
PGND
PGND_2
PGND_3
PGND_4
R182 16.9K
R183 6.04k
R184 4.87k
R185 11.8k
R186 10K
R187 10K
R188 49.9
R189 10
R190 49.9
C282 100uF
C283 10uF
C284 10uF
C285 10uF
C286 0.1uF
C287 2.2uF
C288 100nF
C289 47uF
C290 100uF
C291 100uF
C292 100uF
0.95V_VDDC
U31
TPSM843822RDG
VIN_1
VIN_2
VIN_3
VIN_4
VIN_5
BOOT
SW
VOUT_2
VOUT
FB
GOSNS
BPS
EN
PG
SYNC_FSEL
MSEL
C304 51.27uF
C305 1uF
C306 2.98uF
C307 1uF
C308 12nF
C309 47uF
R200 100k
R201 11.8k
R202 4.02k
R203 56k
R204 10k
U30
TPSM843620SIT
R191 16.9K
R192 6.04k
R193 4.87k
R194 10K
R195 49.9
R196 10
R197 11.8k
R198 10K
R199 49.9
C293 100uF
C294 10uF
C295 10uF
C296 10uF
C297 0.1uF
C298 47uF
C299 100uF
C300 100uF
C301 100uF
C302 100nF
C303 2.2uF
0.95V_VP
R11 1K
R12 330R
R13 330R
DS1 Red
DS3 Red
DS2 Red
R16 5.1k
R17 330R
R18 5.1k
R19 330R
DS4 Red
DS5 Red
Q1 2N7002A
Q2 2N7002A
U2_PGOOD
U3_PGOOD
0.9V_VDDC indicator
0.9V_VP indicator


3.3V_EXT
U4 APT381C-FGE-7
IN
NC3
NC2
EN
OUT
NC1
ADJ
GND
C36
22uF/25V
C37
0.1uF
R25
3.9K1%
R26
20K1%
0.95V_CVDDC
C38
10uF/10V
C39
0.1uF
C40
0.1uF
3.3V_EXT
U5 APT381C-FGE-7
C41
22uF/25V
C42
0.1uF
R27
25K1%
R28
20K1%
1.8V_EXT
C43
0.1uF
C44
0.1uF
C45
10uF/10V